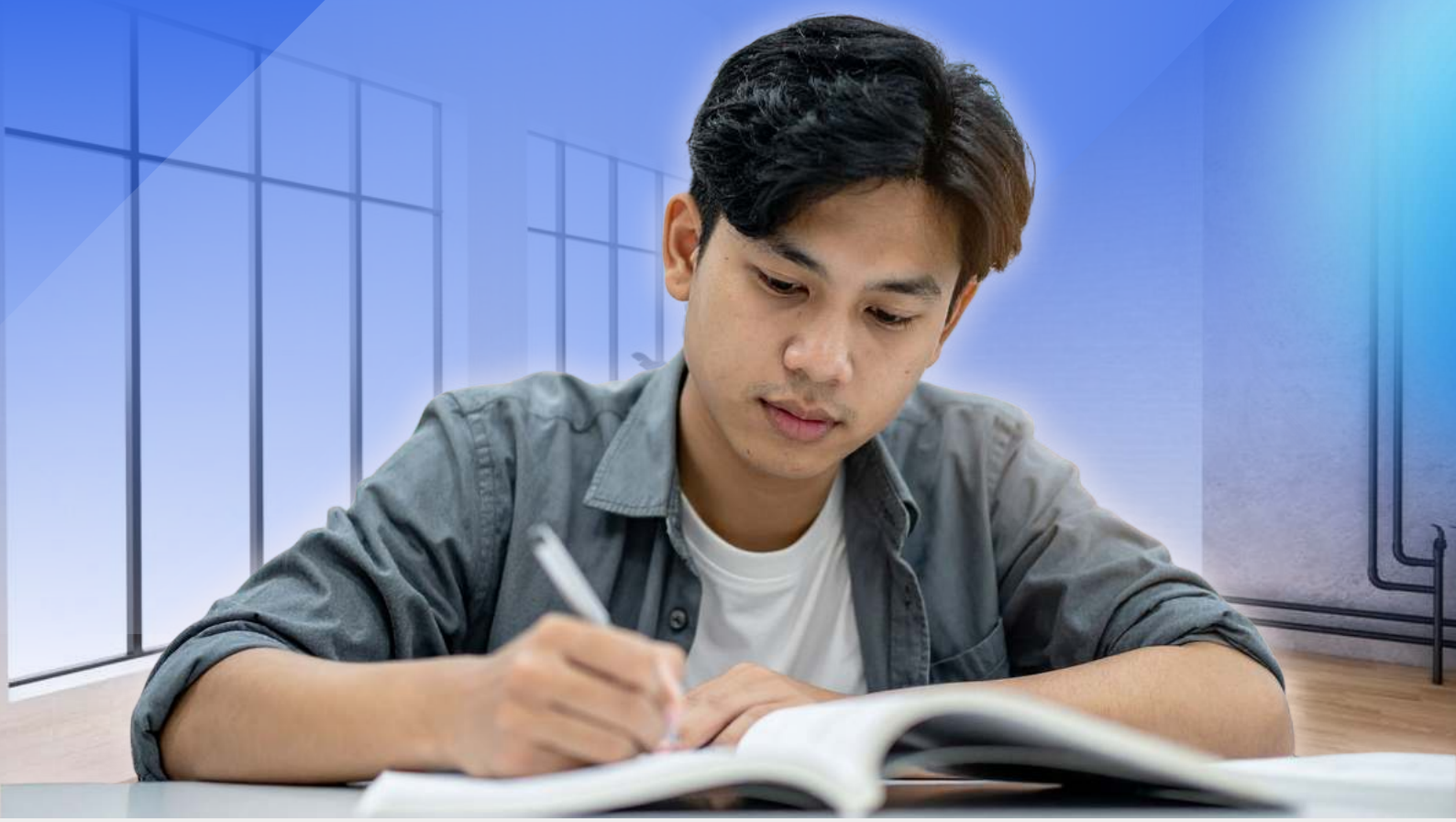
Lenterago

TRY OUT #3

SAT MATH

61004

44 QUESTIONS



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MODUL 1

22 QUESTIONS

DIRECTIONS

The questions in this section address a number of important math skills.
Use of a calculator is permitted for all questions.

NOTES

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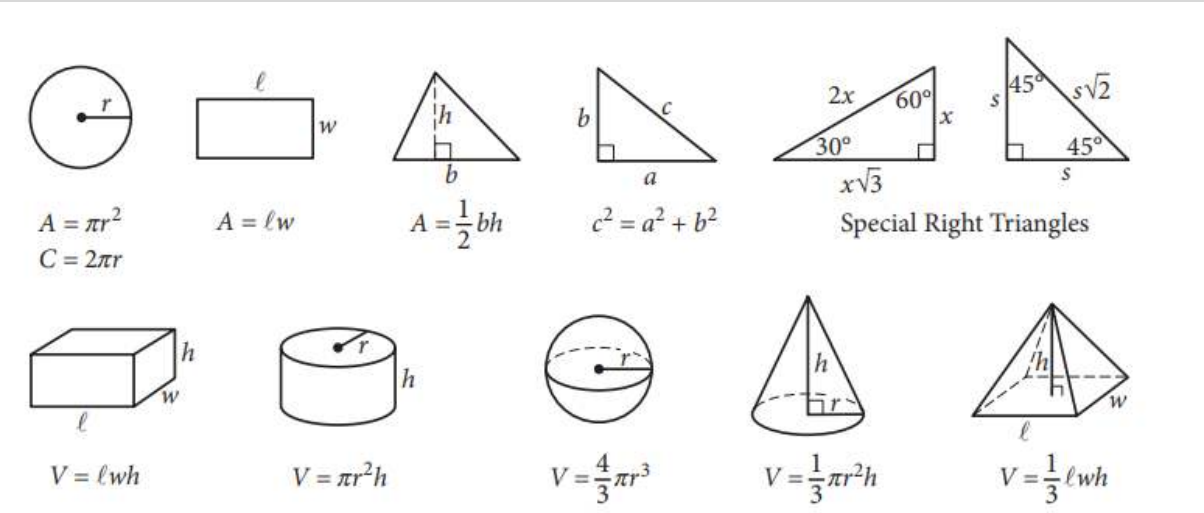

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2p.

The sum of the measures in degrees of the angles of a triangle is 180.

0 8 2 1 7 0 2 7 6 0 7 6

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Module 1

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1. A local café sells gift cards. Leo loads x dollars today, then the cashier adds a bonus of \$12 and removes a \$5 activation fee. If the final balance equals \$3x − 7, what is x?
   A) 4
   B) 6
   C) 8
   D) 10

   Explanation: $x + 12 - 5 = 3x - 7 \Rightarrow x + 7 = 3x - 7 \Rightarrow 14 = 2x \Rightarrow x = 6$.
2. A rectangular garden has width w meters. Its length is 2w + 5 meters. What is the garden's perimeter (meters)?
   A) 4w + 5
   B) 6w + 5
   C) 6w + 10
   D) 8w + 10

3. A charity's "weighted score" is E = (O + 4M + P)/6. Solve for P in terms of E, O, and M.
   A) 6E − O − 4M
   B) 6E + O − 4M
   C) 6E − 4O − M
   D) O + M − 6E

4. In right triangles ABC and DEF, ∠C and ∠F are right angles, and A corresponds to D. If ∠B = 35°, what is ∠D?
   A) 35°
   B) 45°
   C) 55°
   D) 145°

5. A club sells 3-ticket and 5-ticket bundles for a concert. Exactly 50 bundles are sold and they cover 180 seats when full. How many 3-ticket bundles were sold?
   A) 25
   B) 30
   C) 35
   D) 40

6. A work site must place scaffolding at a point (x, y) that lies above the line y = 3x − 2 and strictly to the right of the vertical line x = 4. Which statement must be true?
   A) $y \geq 10$
   B) $y > 10$
   C) $y \geq 14$
   D) $y > 14$

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7. A tank's depth in meters is modeled by $d = \sqrt{(2t + 3)}$. A sensor also reports the surface is 1 meter above the tank's rim when the reading equals t. Solve $\sqrt{(2t + 3)} + 1 = t$.
   A) $1 + \sqrt{6}$
   B) $2 + \sqrt{6}$
   C) 3
   D) no real solution

8. For $x > 2$, simplify the machine-efficiency ratio $(x^3 - 8)/(x^2 - 4)$.
   A) $x + 2$
   B) $(x^2 + 2x + 4)/(x + 2)$
   C) $x - 2$
   D) $(x^2 - 2x + 4)/(x - 2)$

9. A circular pond has equation $(x - 4)^2 + (y + 2)^2 = 25$ in meters. A pier is at (9, k) on the shore. What is k?
   A) −5
   B) −3
   C) −2
   D) 2

10. A delivery van travels x miles on highway and (120 − x) miles on city streets. Fuel used (gallons) is $F = x/30 + (120 - x)/20$. Which is equivalent?
    A) $F = 4 - x/60$
    B) $F = 6 - x/60$
    C) $F = 4 + x/60$
    D) $F = 6 + x/60$

    Explanation: $x/30 + 120/20 - x/20 = x/30 + 6 - x/20 = 6 - x/60$.

11. A parabola has x-intercepts −4 and 1 and y-intercept −8. Which equation fits?
    A) $y = (x + 4)(x - 1)$
    B) $y = 2(x + 4)(x - 1)$
    C) $y = -2(x + 4)(x - 1)$
    D) $y = (x - 4)(x + 1)$

12. A company's output O in week n doubles every week: $O_n = O_1 \cdot 2^{n-1}$. If $O_1 = 150$ and $O_n/O_1 = 1/8$, what is $n - 1$?
    A) −3
    B) −2
    C) 2
    D) 3

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13. A reservoir's water level declines linearly from 6 m to 2.1 m over 15 years because of drought mitigation. Which model (meters, t years after start) best fits?
    A) $f(t) = -0.26t + 6$
    B) $f(t) = -0.39t + 6$
    C) $f(t) = 0.26t + 6$
    D) $f(t) = 0.39t + 2.1$

14. How many intersections does the system have?
    $y = x^2 - 4x + 1$
    $2x + y = 5$
    A) 0
    B) 1
    C) 2
    D) infinitely many

15. A street-lamp pole casts a 9-ft shadow at the same time a 6-ft person casts a 4-ft shadow. Assume sun rays are parallel and triangles similar. Pole's height?
    A) 10.5 ft
    B) 12 ft
    C) 13.5 ft
    D) 15 ft

16. Solve for the positive root: $x^2 - 7x - 18 = 0$.

17. Combine polynomials to get $ax^2 + bx + c$:
    $(-3 + 5x - 2x^2) + (6x^2 - 4x + 1)$. What is $a + b + c$?

18. Two jobs yield:
    $x - 2y = -7$ and $3x + y = 4$.
    Find y?

19. Slope of the line through $(-3, 5)$ and $(2, -1)$, simplified as a fraction.

20. An arc is 3/8 of a circle. Central angle (degrees)?

21. A geometric sequence models the number of downloads: first hour 5, then each hour triples. The nth-hour count is
    A) $5 \cdot 3^n$
    B) 15^n
    C) $3 \cdot 5^{n-1}$
    D) $5 \cdot 3^{n-1}$

22. Two right triangles are similar. One has an acute angle 22°. What is the measure of the corresponding acute angle in the other triangle?
    A) 22°
    B) 38°
    C) 68°
    D) 90°

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MODUL 2

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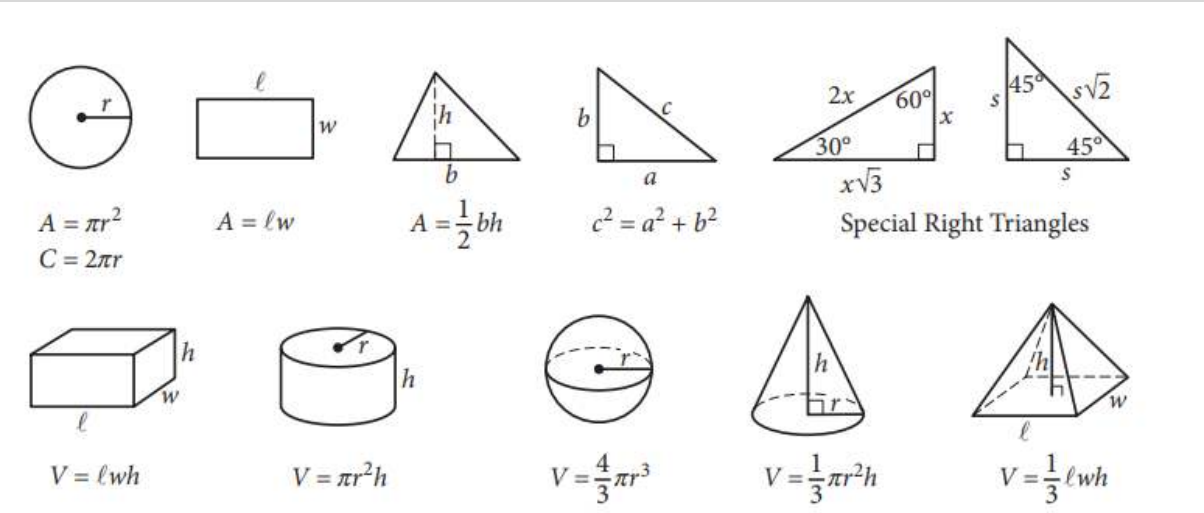

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23. A grocery store advertises apples at $2.50 per pound. Nora buys c pounds and uses a $1 coupon applied once to the total. Her cost is
    A) $2.5c - 1$
    B) $2.5c - c$
    C) $2.5(c - 1)$
    D) $2.5c + 1$

24. Find the slope of $5y - 10x = 15$.
    A) -2
    B) $-1/2$
    C) 2
    D) 5

25. A manufacturer inspects 200 phones; 3 are defective. Assuming the rate holds, how many of 10,000 phones are expected to be defective?
    A) 120
    B) 150
    C) 300
    D) 600

26. A forester models average tree height H (cm) by $H = 1.67A + 21.1$, where A is age in years. Predict the height of a 19-year-old tree.
    A) 49.7
    B) 52.8
    C) 54.9
    D) 57.1

27. Two numbers have sum 26 and difference 6. The larger number is
    A) 10
    B) 13
    C) 16
    D) 20

28. A stadium sold bench seats at $75 each and lawn seats at $40 each. A total of 350 tickets brought in $19,250. Which system models the situation with B bench and L lawn seats?
    A) $B + L = 350$; $75B + 40L = 19{,}250$
    B) $B - L = 350$; $40B + 75L = 19{,}250$
    C) $75B + L = 350$; $B + 40L = 19{,}250$
    D) $40B + 75L = 350$; $B + L = 19{,}250$

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29. Which line has slope 3?
    A) $y = 3x + 2$
    B) $3y = x + 2$
    C) $y = -3x + 2$
    D) $y = 2x + 3$

30. If $x/(x + 1) = 2/(x + 1)$ and $x \neq -1$, then $x =$
    A) -2
    B) 0
    C) 1
    D) 2

31. A right circular cone has volume 150π cm³ and height 12 cm. Its radius r is
    A) 5
    B) $(5\sqrt{6})/2$
    C) 6
    D) 7

32. A water cooler holds 1 gallon = 128 fl oz. How many 16-oz cups can be filled?
    A) 6
    B) 7
    C) 8
    D) 10

33. An insurance agent set a goal of at least 57 policies but sold fewer than that; total coverage value exceeded \$3,000,000. Let x be \$50k policies and y be \$100k policies. Which system fits?
    A) $x + y \geq 57$; $50{,}000x + 100{,}000y > 3{,}000{,}000$
    B) $x + y < 57$; $50{,}000x + 100{,}000y > 3{,}000{,}000$
    C) $x + y > 57$; $50{,}000x + 100{,}000y \leq 3{,}000{,}000$
    D) $x + y < 57$; $50{,}000x + 100{,}000y \leq 3{,}000{,}000$

34. Solve $4^x = 1/64$.
    A) -4
    B) -3
    C) 3
    D) 4

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35. For which x is $(x - 10)/(x^2 - 9)$ undefined?
    A) −10
    B) −3
    C) 3
    D) B and C

36. A rectangular solid measures 30 × 40 × 50 cm and has density 2.8 g/cm³. What is its mass (grams)?
    A) 84,000
    B) 120,000
    C) 168,000
    D) 336,000

37. In a clinical study, 33 of 150 adults on a placebo and 21 of 150 adults on vitamin C reported a cold. What fraction represents the placebo group's proportion?
    A) 7/50
    B) 11/50
    C) 7/25
    D) 11/25

38. A class's ages (years): 18 occurs 6 times; 19 (5); 20 (4); 21 (2); 22 (1); 24 (1); 29 (1). Order mean, median, mode from least to greatest.
    A) mean < median < mode
    B) mode < median < mean
    C) median < mode < mean
    D) mode < mean < median

39. A fireworks height (meters) is modeled by $h(t) = -a(t - 0.5)^2 + 5$ with zeros at $t = -2$ and $t = 3$. What is a?
    A) 1/2
    B) 0.8
    C) 1
    D) 1.2

40. An office retrofit costs $24,000 and reduces yearly energy cost from $4,300 to $2,800. Which inequality gives the number of years t until savings exceed the cost?
    A) 24,000 < 2,800t
    B) 24,000 < 4,300t
    C) 24,000 < (4,300 − 2,800)t
    D) 24,000 < (2,800 − 4,300)t

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41. Plastic production is modeled by y = 3.39x + 46.89 (billions of pounds), x = years since 1985. What does 3.39 represent?
    A) starting output
    B) average yearly increase
    C) data spread
    D) correlation

42. A city's recycling output is modeled by
    $R(t) = 4.2t + 75.5$
    where $R$is measured in thousands of tons and $t$is the number of years since 2015. Approximately what percent increase does the model predict from 2018 ($t = 3$) to 2022 ($t = 7$)?
    A) 16%
    B) 18%
    C) 19%
    D) 22%.

43. Solve $3^{\{x+1\}} = 81$. Find x=?

44. A circular track has radius 10 m. A drone flies along a 72° arc. What is the arc length (meters)?

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